Stat 4210 - Final Project Report

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Introduction

Depression is a loss of pleasure in activity and sense of hopelessness often combined with fatigue, feelings of guilt, loss of interest in activities, and inability to concentrate (American Psychiatric Association, 2013). It affects 13.1% of adults and adolescents in the United States, with a higher prevalence in females than males (CDC Statistics, 2025). Prevalence is significantly higher in college students, with rates climbing to 38-48.9% (Healthy Minds Network, 2024 ;Luo et al., 2024). The goal of this project is to investigate how visible life stressors affect depression among college students, with a focus on factors such as sleep, work, work satisfaction, study time, and study satisfaction. Previous research shows correlations between GPA, depression, and study stress (Luo et al., 2024), with 77% of students with depression reporting academic impairment (Healthy Minds Network, 2024). Similarly, sleep disruption is linked to a number of psychiatric disorders, including major depressive disorder (Sun et al., 2022; Chunnan et al., 2022), while poor work-life balance has also been shown to correlate with higher depression rates in college students (Sprung & Rogers, 2021). These relationships are likely bidirectional: stressors such as poor sleep or high academic pressure may contribute to depression, while depression may also worsen these stressors. This feedback loop of increasing stress and depression shows the need for predictive models that can identify at-risk students based on observable factors, no matter which factor came first.

Dataset & Origin

Dataset Overview

The Student Depression Dataset (Opeyemi, 2024) is sourced from Kaggle, a leading platform for data science and machine learning. This specific set was compiled from anonymized survey data across several different schools and contains approximately 27,900 student records, where each row represents a single student.The dataset is designed to study and predict depression among students by combining demographic, academic, lifestyle, and wellbeing-related variables. It is a great resource for the project because it takes many factors into consideration, things like how much they are sleeping, financial stress and how much pressure they feel from school.

Importance

This dataset contains data from a large number of individuals, providing enough information to run complex models without worrying about small sample sizes. Prediction in this context is extremely significant because it can help identify at-risk students early based on observable academic, lifestyle, and demographic factors, allowing counselors and faculty to make timely mental health interventions and provide support. These insights will help schools identify high risk lifestyle patterns, allowing schools to improve their mental health resources on campus.

Available Predictors

Table 1 lists all of the available predictors for the Student Depression Dataset (Opeyemi, 2024). Not all predictors were used for every model. Table 2 specifies the predictors used in each model with more detailed explanations for eliminating variables in the sections specific to each model.

Table 1 Available predictors from Student Depression Dataset

Variable	**Description**	**Type**
Gender	The gender of the student (Male/Female)	Binary
Age	The age of the student (years)	Continuous
City	The city or region where the student resides (city name)	Multi-class
Academic Pressure	A measure indicating the level of pressure the student faces in academic settings (integer levels 1-5)	Ordinal: multi-class
Work Pressure	A measure of the pressure related to work or job responsibilities (integer levels 0-5)	Ordinal: multi-class
Study Satisfaction	An indicator of how satisfied the student is with their studies (integer levels 1-5)	Ordinal: multi-class

Job Satisfaction	A measure of the student's satisfaction with their job or work environment (integer levels 1-5)	Ordinal: multi-class
Sleep Duration	The average number of hours the student sleeps per day (range survey response)	Ordinal: multi-class
Dietary Habits	An assessment of the student's eating patterns and nutritional habits (Healthy/Moderate/Unhealthy)	Multi-class
Degree	The academic degree or program that the student is pursuing	Multi-class
Have you ever had suicidal thoughts?	A binary indicator (Yes/No) that reflects whether the student has ever experienced suicidal ideation.	Binary
Work/Study Hours	The average number of hours per day the student dedicates to work or study (hours)	Continuous
Financial Stress	A measure of the stress experienced due to financial concerns (integer levels 1-5)	Ordinal: multi-class
Family History of Mental Illness	Indicates whether there is a family history of mental illness	Binary
Depression	Whether the student is experiencing depression (Yes/No)	Binary

Topics/Models Included

Table 2 summarizes each model and shrinkage and fitting applied to each model and lists the final variables used in each model after the shrinkage and/or fitting was applied.

Table 2 Model Fitting Methods and Final Variables

Model	**Shrinkage Methods/Fitting**	**X Input Variables**	**Y Dependent Variables**
Linear Regression	Lasso, Ridge, Elastic Net	Age, Academic Pressure, Study Satisfaction, Sleep Duration, Dietary Habits, Work/Study Hours, Financial Stress, Family History of Mental Illness, CGPA,“Have you ever had suicidal thoughts?”	Depression
Linear Regression	Lasso, Ridge, Elastic Net	Age, Academic Pressure, Study Satisfaction, Sleep Duration, Dietary Habits, Work/Study Hours, Financial Stress, Family History of Mental Illness, depression, “Have you ever had suicidal thoughts?”	CGPA
Logistic Regression	Lasso (with hyperparameter tuning)	Uses all variables for Standard, and removes “Have you ever had suicidal thoughts?” for the sparse model(s)	Depression
Discriminant Analysis	Grid search for optimal regularization parameter	Includes all variables except “Have you ever had suicidal thoughts?” and “Depression”	Depression

K-Nearest Neighbors	Number of neighbors (k) and Distance Weighting	Includes all variables except “Have you ever had suicidal thoughts?” and “Depression”	Depression
Deep Learning	Early stopping, Dropout, weight decay	Includes all variables except “Have you ever had suicidal thoughts?” and “Depression”	Depression
General Additive Model	Grid search for optimal lambda (smoothing parameter)	Includes all variables except “Have you ever had suicidal thoughts?” and “Depression”	Depression
Tree Based Method	Gini Impurity, Bootstrap Aggregation, and learning rate	Includes all variables except “Have you ever had suicidal thoughts?” and “Depression”	Depression
Support Vector Machine	L_2 Penalty, C Hyperparameter, and Kernel Fitting	Includes all variables except “Have you ever had suicidal thoughts?” and “Depression”	Depression

These models were primarily used for binary classification, where the goal was to predict whether a student was experiencing depression.

1. Linear Regression

Professor Comment: “need to do this and use plenty of predictors”

A multiple linear regression model was first used to fit the Student Depression data. The main assumptions of the linear regression are: variables have a linear relationship; observations are independent; residuals have constant variance; error distribution is normal; and multicollinearity is absent. The linear regression model used abides by these assumptions.

1.1 Ordinary Least Squares (OLS) Model The ordinary least squares model was used for this multiple linear regression. In an ordinary least squares model, the best fit line is drawn by minimizing the sum of squares, or residuals, between observed value and predicted value for each data point.

1.1.1 Qualitative Predictors Qualitative predictor variables were turned into “dummy variables”, so that they could be included in the linear regression analysis. These included sleep duration, dietary habits, “Have you ever had suicidal thoughts?”, and family history of mental illness. However, qualitative variables that did not have an apparent rank could not be turned into meaningful “dummy variables” and therefore were excluded from the linear regression model. These included city, gender, profession, and degree.

1.1.2 Coefficient Selection After “dummy variables” were created, coefficients, standard error, t-statistic, and p-values were generated. Predictor variables with p-values less than 0.05 were removed from the dataset because they were not significant predictors of depression. Predictors removed were job satisfaction (p=0.682) and work pressure (p=0.700). The predictor sleep duration [7-8 hours] (p=0.449) was kept as it was part of a category of dummy variables and removing it would change the reference point. Confidence intervals and prediction intervals were generated for the remaining coefficients.

1.1.3 Multicolinearity Then, multicollinearity was assessed using variance inflation factor (VIF). Multicollinearity, which is high correlation between predictor variables, can lead to unstable coefficients, and therefore, those variables should be combined or removed from the dataset to limit damage. VIFs between 5 and 10 generally indicate collinearity. All VIFs in the Student Depression dataset were below 3, suggesting that the predictor variables were not correlated. The intercept had a VIF of 87, but intercepts are expected to have high VIFs, so this was not a problem. Therefore, no predictor variables were removed or combined due to collinearity.

1.1.4 Residual analysis Finally, residuals were analyzed using a residuals versus fitted values plot and a Q-Q plot. The residuals versus fitted values plot took the form of a funnel, indicating heteroscedasticity, which means that the residuals are not constant as the model fit increases, violating the constant variance assumption. To address this problem, a robust covariance estimation, which minimizes the influence of outliers, was applied. Transforming the data logarithmically or exponentially was not an option to address the variance because of the binary variables. The Q-Q plot generated a large tail, deviating from the best fit line on both sides. This indicates that the residuals were higher than predicted.

1.1.5 Cross Validation Cross validation, testing the model on a subset of the data instead of the entire dataset, was performed to ensure that the model performed nearly as well on test data as it did on training data. The model performed nearly as well on the testing data as it did on the training data, so further analysis continued.

1.2 Shrinkage methods Shrinkage was applied by ridge regression, lasso regression, and elastic net to obtain the most accurate model. Shrinkage methods apply a penalty to coefficients to decrease the complexity of the model and help limit overfitting. This is beneficial with data sets with large numbers of predictors where eliminating coefficients makes the models easier to interpret. Elastic net combines both types of penalties. Elastic net is beneficial

when data sets have more predictors than observations or when data sets have high multicollinearity. To determine the ideal coefficients for each model, mean squared error was plotted against penalty size and the penalty size resulting in the lowest prediction error was selected. Coefficients and coefficient shrinkage plots were generated.

1.2.1 Comparison of linear regression models The shrinkage models were compared to the regular multiple linear regression model using the metrics R squared, adjusted R squared, mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE). These metrics were the same for the original model and all three shrinkage models. MSE, RMSE, and MAE were all low, indicating that the model has low error. However, R squared and adjusted R squared were both low, indicating that the predictor variables do not do a great job explaining the outcome variable, most likely due to the binary predictors and the “dummy variables”. Mallow’s Cp was used to further evaluate the standard error. Mallow’s Cp compares total mean square error to number of parameters to determine model accuracy. In addition, AIC, BIC, and process capability index (Cp) were also used to compare the models. Lower Cp indices have less error and therefore indicate greater model accuracy. The ridge regression model has the lowest Cp value and therefore greatest accuracy. AIC and BIC compare the goodness of fit and complexity of models, penalizing models for complexity. BIC applies a harsher penalty for the number of parameters, which increases with sample size, than does AIC. For these metrics, lower values are usually preferred and indicate less complex models that are still accurate. The ridge regression model had a slightly lower value than lasso, elastic net, and ordinary least squares models. Therefore, it is the better model to use. It should be noted that the metrics were close between models and ridge regression barely performed better than the other models. Due to the number of qualitative predictors, logistic regression would likely provide a better model for the Student Depression dataset and will be examined in the next section.

1.3 CGPA as outcome variable As linear regression is a prediction method, using a continuous variable as the outcome variable would be most appropriate. In the previous linear regression model, the outcome variable, depression, was binary, which is better suited for classification models. Therefore, we reran the linear regression model using CGPA, the only continuous variable in the dataset, as the outcome variable. Surprisingly, the model with CGPA as the outcome variable performed considerably worse than the previous model. The R-squared for the OLS when CGPA was the outcome variable was 0.0020. The shrinkage methods - ridge, lasso, and elastic net - were also all 0, while the R-squared with depression as the outcome variable were all 0.5 for the OLS model and all shrinkage models. Further, the MSE, RMSE, and MAE were all higher for the model with CGPA as the outcome variable, indicating that it had greater error. In addition, the BIC, AIC, and Cp values of the model with CGPA as the outcome variable were all of higher magnitude indicating worse performance.

Depression Models — AIC, BIC & Mallows' Cp

	AIC	BIC	Cp
OLS	-59,734.21	-59,925.73	14.00
Ridge	-59,736.19	-59,629.13	12.02
Lasso	-59,735.99	-59,628.93	12.22
Elastic Net	-59,735.99	-59,628.93	12.22

CGPA Models — AIC, BIC & Mallows' Cp

	AIC	BIC	Cp
OLS	100,515.49	100,622.55	13.00
Ridge	100,517.26	100,624.32	14.77
Lasso	100,510.26	100,576.14	7.76
Elastic Net	100,510.26	100,576.14	7.76

2. Logistic Regression

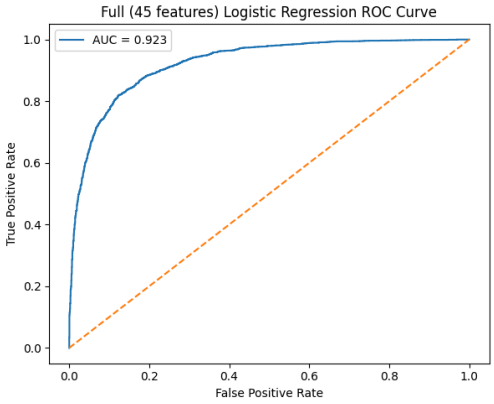
Professor Comment: “good”

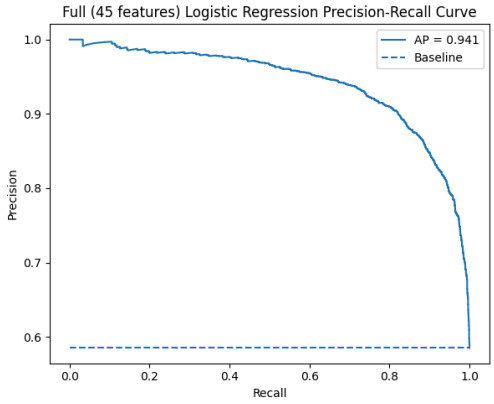
Logistic regression was selected as the primary classification model for the Student Depression dataset because the response variable, Depression (Yes/No), is binary. Unlike linear regression, which predicts a continuous response, logistic regression predicts the log-odds of an event occurring, and maps that output to a probability between 0 and 1 using the sigmoid function. This makes it well-suited for binary classification, particularly when interpretability and inference are important.

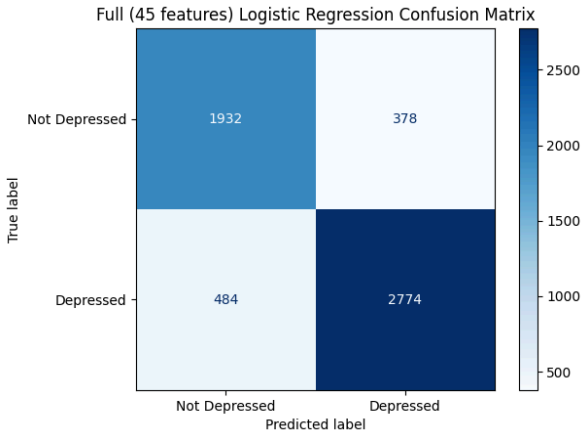
Response variable (Y): Depression (binary: 0 = Not Depressed, 1 = Depressed)
Predictor variables (X): Age, Academic Pressure, CGPA, Study Satisfaction, Work/Study Hours, Financial Stress, and four interaction terms (Academic Pressure × Sleep Duration, Academic Pressure × Financial Stress, Financial Stress × Sleep Duration, Academic Pressure × Study Satisfaction), along with one-hot encoded categorical variables: Gender, Sleep Duration, Dietary Habits, Degree, Have you ever had suicidal thoughts?, and Family History of Mental Illness.

2.1 Standard Logistic Regression

A baseline logistic regression model was fit using all available predictors after preprocessing. Because the dataset exhibits some class imbalance, class_weight='balanced' was used, which weights the classes based on their frequency, minimizing the effect of the unbalanced data. Five-fold cross-validation was applied to the training set to evaluate generalization, and to identify any unexpected behavior after training. Classification performance was then evaluated on the held-out test set (20% of data) using our full array of evaluation metrics (ROC-AUC, accuracy, precision, recall, F1 score, MCC, etc.)







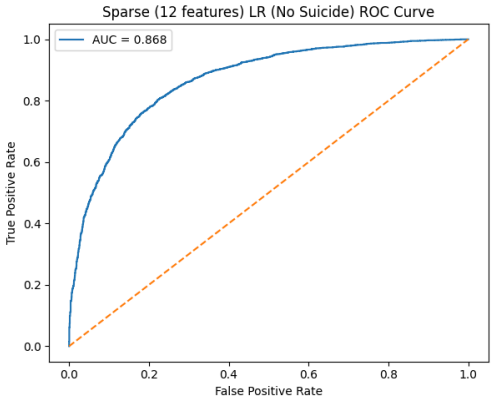
2.2 Sparse Logistic Regression (Lasso)
To improve interpretability and reduce model complexity, a Lasso penalty was applied to the logistic regression. The Lasso penalty drives some coefficients exactly to zero, effectively doing automatic feature selection. This is useful because of our one-hot encoded categorical variables, which introduce many binary columns with little effect on the model.

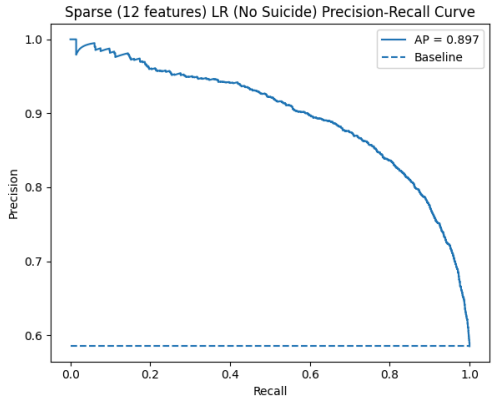
The regularization strength is controlled by the hyperparameter C (the inverse of λ; lower C means stronger regularization). C was tuned using a grid search over 50 values, with 5-fold cross-validation of ROC-AUC as the score. Rather than simply selecting the C value that maximized ROC-AUC, the simplest model whose score fell within one standard deviation of the best score was selected. This "one-standard-error rule" allowed us to choose a parsimonious model while maintaining reasonable accuracy.

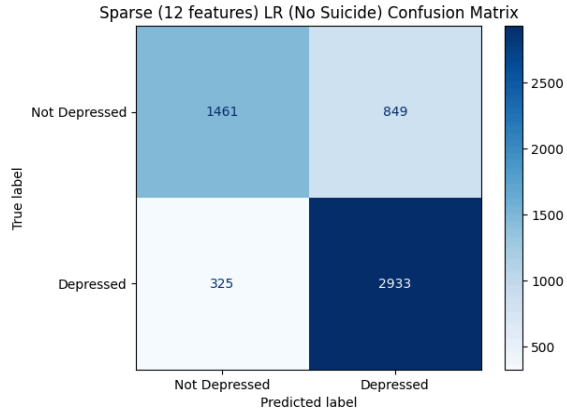
Because missing a depressed student (a false negative) carries a higher real-world cost than a false alarm, the classification threshold was lowered below the default 0.5. Specifically, the threshold was tuned to achieve a minimum recall of 90% on the training set, and the best-precision threshold for that constraint was selected.

2.3 Logistic Regression Without Suicidal Ideation (Primary Model)
The variable “Have you ever had suicidal thoughts?” is closely intertwined with a clinical depression diagnosis, as suicidal ideation is itself a major diagnostic result. Including it as a predictor risks data leakage, as the model may achieve inflated accuracy simply by hyper-focusing on this one variable rather than learning the broader lifestyle and academic risk factors the project is trying to study. For this reason, this variable was removed and the full L1-regularized logistic regression pipeline was re-run on the reduced feature set. This version is considered the primary logistic regression model for the purposes of comparison with other methods.

The same hyperparameter tuning procedure (five-fold CV over 50 values of C, one-standard-error rule) and threshold adjustment (target recall ≥ 0.90) were applied identically. Feature importance was assessed by examining the magnitude and sign of the final non-zero coefficients. Because all predictors were standardized prior to fitting, coefficients are on a comparable scale and their absolute values directly reflect each predictor's relative contribution to the chance of depression.







3. Discriminant Analysis

Professor Comment: “good”

3.1 Data Pre-Processing and Leakage Prevention

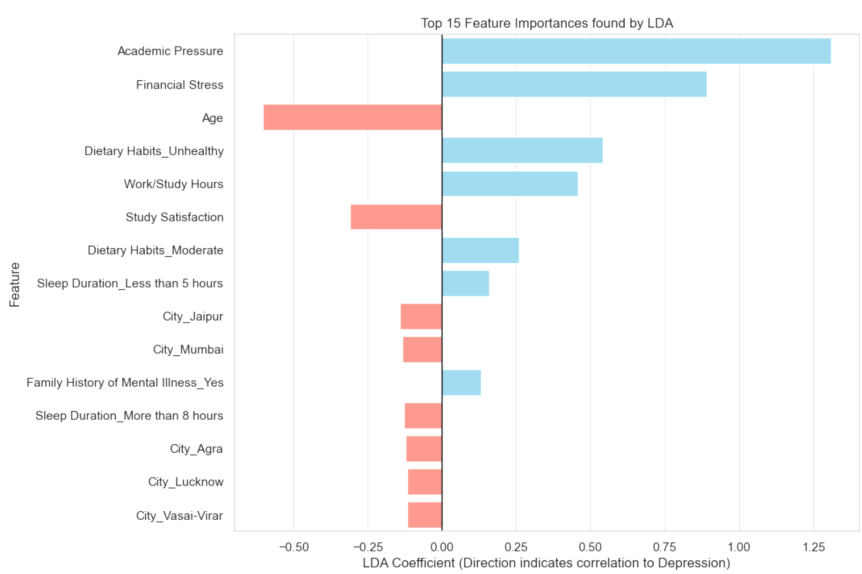
We prepared the dataset by first removing non-predictive identifiers (like id) and the Have you ever had suicidal thoughts? variable. Dropping the suicidal thoughts feature was critical to prevent **Data Leakage**, as retaining it would artificially inflate the model's accuracy and hide the true, actionable causes of depression. Finally, we imputed missing values (median for numerical, mode for categorical), applied One-Hot Encoding to the categorical predictors, and standardized all features using StandardScaler so they contributed equally to the LDA and QDA distance algorithms.

3.2 *Linear Discriminant Analysis (LDA) Implementation and Feature Importance*

The first model implemented was Linear Discriminant Analysis (LDA). LDA assumes that all classes share identical covariance matrices and creates a strictly linear decision boundary. Because of this linear nature, the coefficients calculated by the LDA discriminant natively double as an excellent measure of feature importance. We extracted these coefficients, calculated their absolute magnitudes, and plotted the top 15 determining factors. This allowed us to quantitatively rank exactly which specific student behaviors and stressors drive the model's decision to classify a student as depressed.

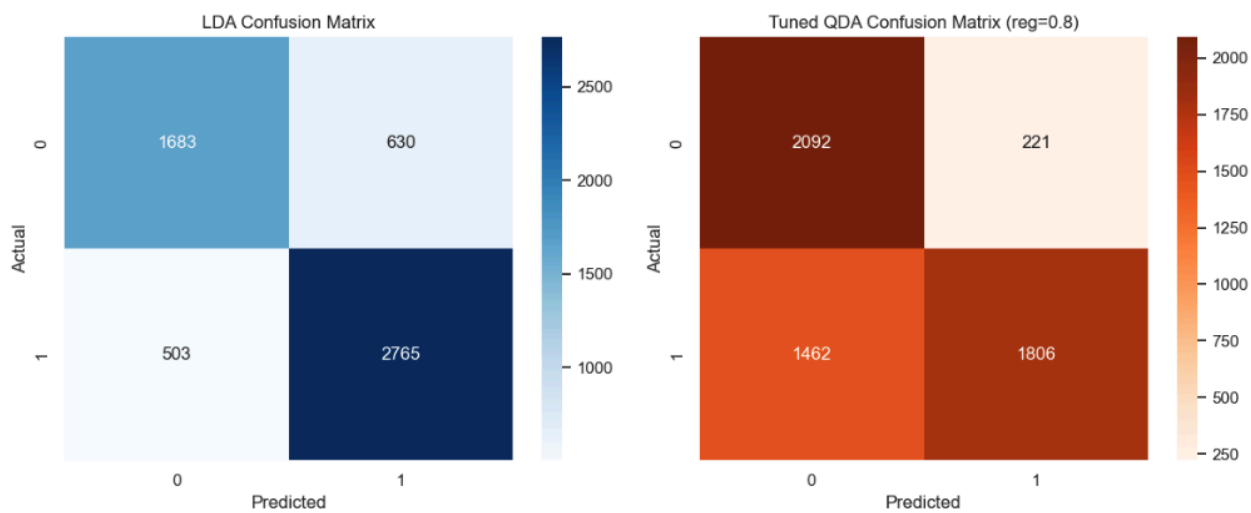
3.3 *LDA Dimensionality Reduction and Subspace Visualization*

Because the target variable (Depression) is strictly binary, the LDA algorithm mathematically compresses the entire multi-dimensional feature space down onto a single 1-Dimensional vector (or line) that maximizes the separation between the two classes. To evaluate the true quality of the dataset, we extracted this 1D projection and plotted the distributions of Depressed vs. Non-Depressed students over it. By visualizing the overlap between the two histograms in this projected subspace, we were able to directly observe where the LDA model was highly confident in its classifications, and where the student populations were too fundamentally similar to separate with a straight line.



3.4 *Quadratic Discriminant Analysis (QDA) and Hyperparameter Tuning*
To address the potential limitations of LDA's linear boundary, we implemented Quadratic Discriminant Analysis (QDA), allowing for curved decision boundaries. Because estimating 108 distinct covariance matrices is highly susceptible to collinearity, we utilized a GridSearchCV with 5-Fold Cross-Validation to mathematically find the optimal shrinkage. The grid search identified reg_param = 0.8 as the optimal regularization parameter to stabilize the matrices.

3.5 Model comparison via Confusion matrices
Despite optimal tuning, the Regularized QDA model achieved a final Accuracy of 69.84%, underperforming the baseline LDA model (79.70%). By plotting the side-by-side visual Confusion Matrices, we observed the structural cause: the extreme sparsity of the 108 One-Hot Encoded categorical variables caused the complex, curved boundaries of QDA to overfit the training data. We conclude that for this specific high-dimensional student dataset, the rigid, linear decision boundaries of LDA provide a vastly superior and more generalized predictive model than QDA.



4. K-Nearest Neighbors

Professor Comment: “good”

We used a K-nearest neighbor model to catch the non-linearity in the dataset and find the more complex patterns. Unlike other models (logistic regression or Neural Networks), it doesn’t utilize a mathematical function or weights during the training. Instead, it attempts to completely memorize the entirety of the training dataset by looking at the *k* most similar students and making a classification based on the majority vote.

4.1 Feature Scaling

One of the most important steps to do when using a KNN model is to scale the numerical data before running. If you do not, the variables that possess larger ranges will take major priority in the model and skew the results. For example, the difference between 5 hours of sleep duration and 5 hours Work/Study hours would be completely different. To fix this, you have to scale the data so it treats a difference of 5 different across all the variables. Furthermore, use the StandardScaler function, which gives the predictors a mean of 0 and a standard deviation of 1.

4.2 Handling Categorical Variables

In order to determine the “closest” students in the dataset, we have to find the distance. However, there is a problem with that. We are unable to find the distance between, say “Male and “Female” or “Engineering” and “Business”. To fix this, convert all of the categorical predictors into numbers using one-hot encoding (creating dummy variables).

4.3 The Curse of Dimensionality

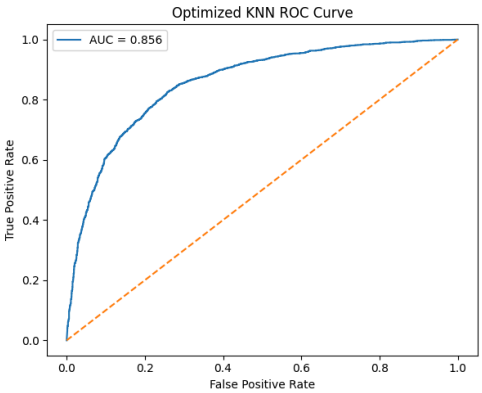
When there are many features, this creates a high-dimensional space. In these cases, KNN struggles and the concept of “distance” falls apart because all of the other data is equidistant from each other. To fix this, use a dimensionality reduction technique like Principal Component Analysis.

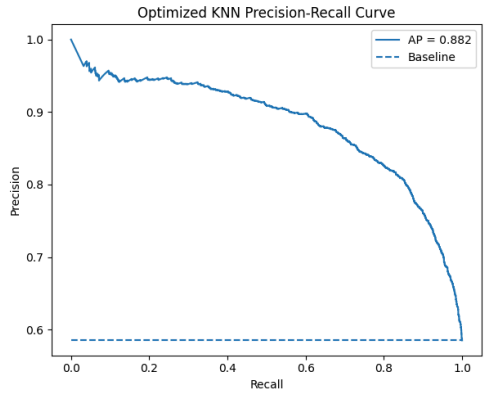
4.4 Hyperparameter Optimization

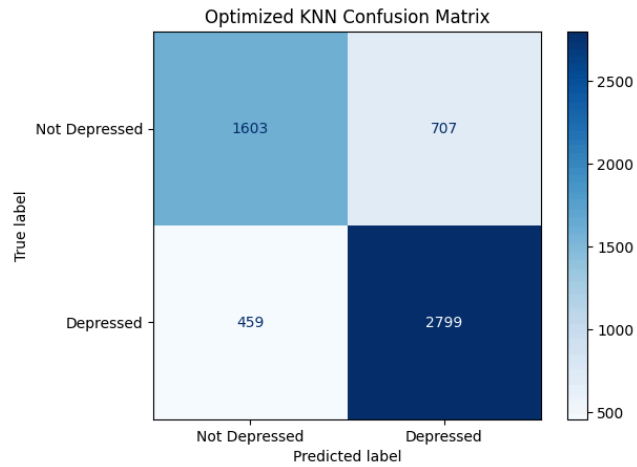
For optimizing the hyperparameters we used the technique GridSearchCV, which will test all the combinations between the three main hyperparameters:

- Number of Neighbors (k):
    - Small k (e.g. 1 or 3): Causes high variance but low bias. Highly sensitive to noise and outliers. The decision boundary is very jagged.
    - Large k (e.g. 25 or 50): Causes low variance but high bias. Less sensitive to noise and smoother but can oversimplify the boundaries.
- The Distance Metric:
    - Euclidean(L_2): This is the straight line distance between the data and is best for continuous data.
    - Manhattan(L_1): Calculates the distance similarly to moving grid-like steps, by using the sum of absolute differences.
- Weighting Function:
    - Uniform: neighbors are all worth the same (1 vote) regardless of distance
    - Distance: The closer the neighbor is to the new student the heavier of a vote it gets. Conversely, the farther the neighbor is, the less of a weight it has.

The hyperparameters that were chosen were **k=41** neighbors **metric=euclidean**, and **weights=distance**. In order to minimize the Expected Prediction Error (EPE), we used 5-fold Cross-Validation. We compared the misclassification error rate across all of the different values of k, and found the value of k that optimized the Bias-Variance Tradeoff.







5. Deep Learning

Professor Comment: “Be sure to use these”

A Multi-Layer Perceptron (MLP) neural network was implemented as the deep learning model for this project. The MLP is a feedforward network in which information flows in one direction from the input layer, through hidden layers, to the output, with no cycles. Each hidden layer applies a linear transformation followed by a non-linear activation function, allowing the network to learn complex, non-linear relationships that simpler models such as logistic regression cannot capture.

5.1 Model Architecture

The network consists of:

- An input layer with dimension equal to the number of predictors
- Two hidden layers with 128 and 64 neurons respectively, each using a ReLU activation and a Dropout layer (rate = 0.3, tuned via validation set testing)
- A single output neuron with a sigmoid activation, giving a predicted probability of depression

ReLU was chosen as the hidden-layer activation function because it is computationally efficient and mitigates the vanishing gradient problem common with sigmoid activations in deep networks. Dropout acts as a regularizer, preventing the network from over-relying on any individual feature, reducing overfitting.

5.2 Training Procedure

The preprocessed data was converted to PyTorch float tensors. The training set was further split into a training partition (80%) and a validation partition (20%), giving three separate sets: training (~56%), validation (~14%), and test (~20%). The validation set was used exclusively during training to monitor loss and trigger early stopping; the test set was held out entirely until final evaluation.

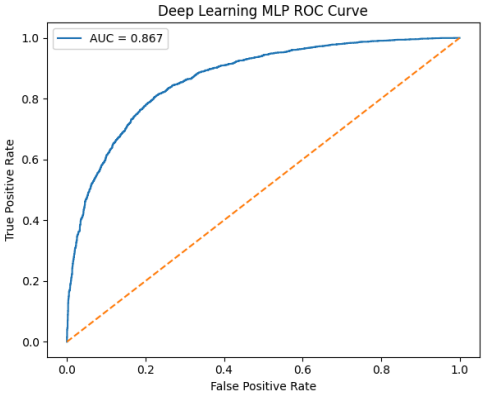
The model was optimized using Adaptive Moment Estimation (Adam) with a learning rate of 0.001 and L2 weight decay of 0.00001. The loss function we used was Binary Cross-Entropy (BCE) Loss, which is the standard for binary classification, equivalent to the log-loss used in logistic regression. Data was fed in batches of 64 samples, with the training set shuffled each epoch.

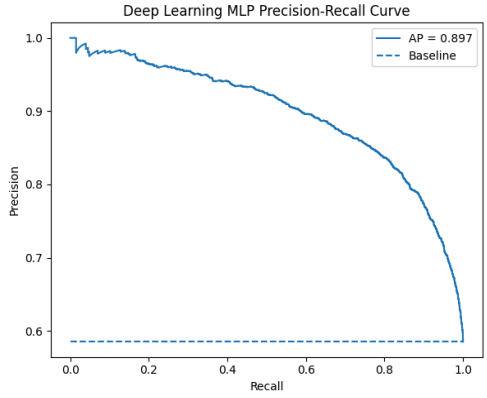
A ReduceLROnPlateau learning rate scheduler was used. If validation loss did not improve for two consecutive epochs, the learning rate was halved. This allowed the optimizer to take larger steps early in training and finer steps as it approached a minimum.

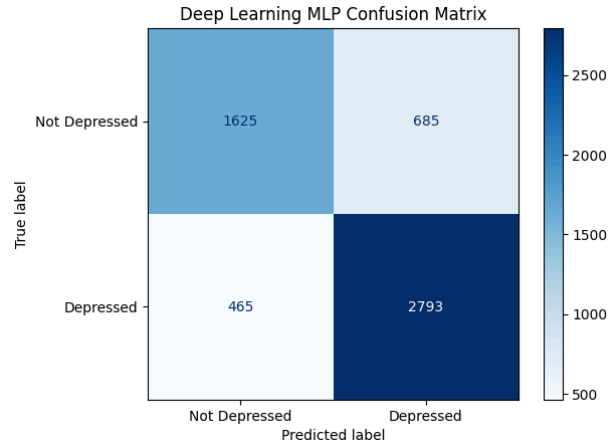
Early stopping was used to prevent overfitting. Training was halted when validation loss failed to improve for 10 consecutive epochs, and the model weights from the best-performing epoch (lowest validation loss) were restored.

5.3 Model Selection

No grid search was conducted for hyperparameters due to computational cost. Instead, the hidden layer sizes (128 and 64), dropout rate (0.30), learning rate (0.001), and weight decay (1×10^{-5}) were set based on common best-practice starting points for data of this size, and tested to give smooth training curves. Cross-validation is not applied in the same fashion as for scikit-learn models because early stopping on the validation set effectively serves the same purpose, and the dataset is large enough (~22,000 training samples) that a fixed train/val/test split gives a good estimate of generalized performance.







6. Resampling

Professor Comment: “cross-validation, validation sets, bootstrap; need to do all except bootstrap”

Resampling methods were used throughout the project to ensure reliable evaluations of model performance and to tune hyperparameters without overfitting to the training data. Five-fold cross-validation was the primary resampling technique, used across logistic regression, discriminant analysis, KNN, GAM, PCR, and PLS to evaluate generalization by rotating which 20% of the training data served as a validation fold. For the deep learning MLP, a dedicated validation set (20% of the training data) was separated out to monitor loss and trigger early stopping, which serves the same purpose as cross-validation for neural networks where re-training is computationally expensive. Weighted splitting was used throughout to preserve the class balance of the depression result across all folds and splits, which was important given the class imbalance in the dataset.

7. Shrinkage Methods

Professor Comment: “LASSO, Ridge; good”

Shrinkage methods were applied in both the linear and logistic regression models to reduce model complexity and avoid overfitting. In the linear regression, ridge (L2), lasso (L1), and elastic net penalties were each used and compared using MSE, AIC, BIC, and Mallow's Cp. Ridge regression produced the lowest Cp and was selected as the best-performing linear model. In logistic regression, a lasso penalty was chosen specifically because its ability to drive coefficients exactly to zero provided automatic feature selection with the many one-hot encoded categorical variables. The regularization strength, controlled by hyperparameter C (the inverse of the penalty weight), was tuned via a grid search over 50 values with five-fold cross-validation, and the one-standard-error rule was applied to select the simplest model within one standard deviation of the best AUC, preferring parsimony over small accuracy gains.

8. Dimension Reduction Methods

Professor Comment: “PCR, PLS; need to do this and use plenty of predictors”

8. Principal Component Regression (PCR) and Partial Least Squares (PLS)

8.1 Feature Expansion

To fully test the capabilities of our dimension reduction models, we needed a large number of predictors. We used PolynomialFeatures to square our continuous variables and create interaction terms (like multiplying Age by CGPA). Combined with our One-Hot Encoded categorical variables, this artificially expanded our dataset to 61 distinct predictors.

8.2 Principal Component Regression (PCR)

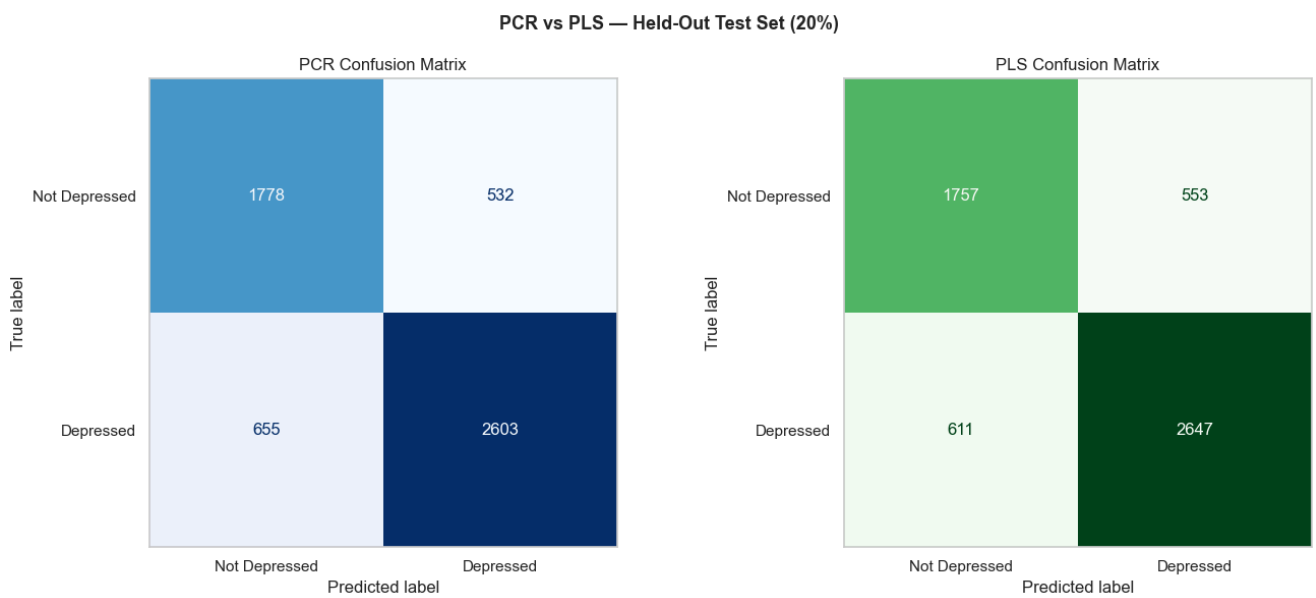
For our first model, we built a PCR pipeline. PCR is simply a two-step process: we first used Principal Component Analysis (PCA) to compress our 61 predictors into a smaller number of components that capture 95% of the total variance in the dataset. We then used only these compressed components to train a Logistic Regression model. Evaluated on the test set, our PCR model achieved an Accuracy of 78.68% and an ROC-AUC of 0.8625.

8.3 Partial Least Squares (PLS-DA)

We then tested Partial Least Squares (PLS) to see if we could improve performance. While PCA compresses data "blindly" (without looking at the target), PLS compresses the features by actively looking for components that have the highest correlation with our target variable (Depression). Because of this targeted compression, our PLS model performed slightly better than the PCR model, achieving an Accuracy of 79.09% and an ROC-AUC of 0.8669.

8.4 Model Comparison via Confusion Matrices

Plotting the cross-validated confusion matrices side-by-side visually confirmed the difference between the two dimension reduction methods. The PLS model successfully identified a higher number of actually depressed students (True Positives) compared to the PCR model. This proves that for this dataset, incorporating the target variable into the dimensionality reduction step (PLS) creates a structurally stronger predictive model than unsupervised compression (PCR).



9. Splines, Local Regression, Generalized Additive Models

Professor Comment: “Non-parametric, non-linear regression models; need to use these, especially generalized additive models with multiple predictors”

9.1 Generalized Additive models and Regression splines

To capture complex, non-linear relationships that strict linear models miss, we implemented a Generalized Additive Model (GAM). We isolated our continuous predictors (e.g., CGPA, Age, Academic Pressure) and passed them through a SplineTransformer utilizing Cubic B-Splines (degree=3) with 4 knots. This transformation mathematically expanded each continuous feature into highly flexible polynomial chunks, allowing the model to fit continuous curved relationships over the data natively.

9.2 Model Architecture and Classification Performance

These transformed splines were combined with One-Hot Encoded categorical predictors and fed into a Logistic Classifier using balanced class weights. The resulting GAM architecture achieved an Overall Accuracy of 79.11% on the unseen test set. Notably, it achieved a Recall of 81% for the Depression class, proving that the flexible spline boundaries were highly effective at capturing True Positives (at-risk students) for clinical screening.

9.3 Validation via Stratified Cross-Validation

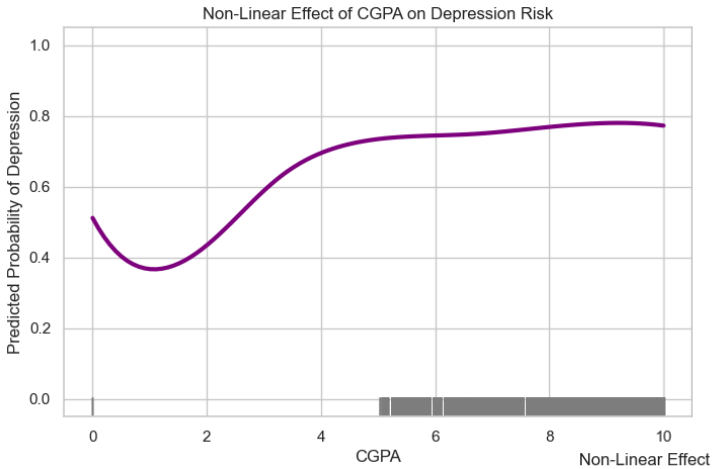
To rigorously prove that the complex spline boundaries were generalizing well and not overfitting, we implemented 5-Fold Stratified Cross-Validation. The model achieved a highly stable Mean 5-Fold ROC-AUC of 0.8733 (± 0.0066). This extremely low variance mathematically guarantees that the underlying polynomial curves successfully capture population-level behavioral trends rather than randomized sample noise.

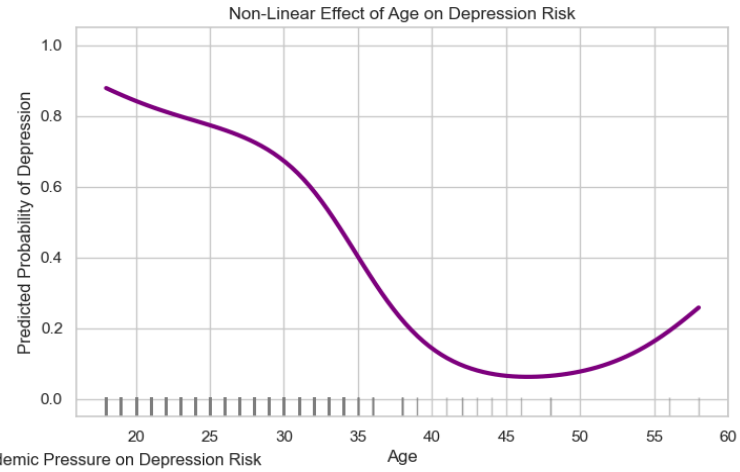
9.4 Interpretation of Non-Linear Curves (Partial Dependence)

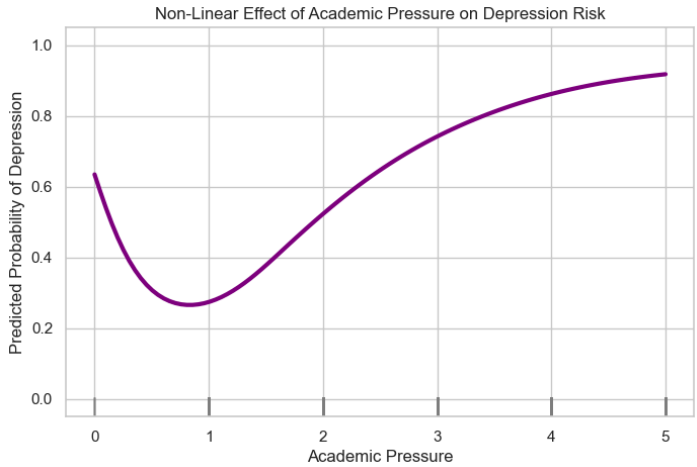
Plotting the model's predicted probabilities across individual features revealed the exact non-linear forces driving student depression:

The U-Shaped CGPA Curve: Depression probability does not perfectly scale with academic failure. While failing students exhibited severe risk (>80%), the curve dipped to its lowest point for average students before curving decisively upwards again for extreme high-achievers (capturing perfectionism-driven stress).

Age Trajectories: The spline boundary mapped a massive risk peak for the youngest demographic (18-22 years old), which smoothly declined throughout their 30s before displaying a slight upward curl as students approached late middle-age. These distinct U-shaped and exponential curves provide concrete statistical proof that the relationships driving mental health risk in this cohort are intrinsically non-linear.

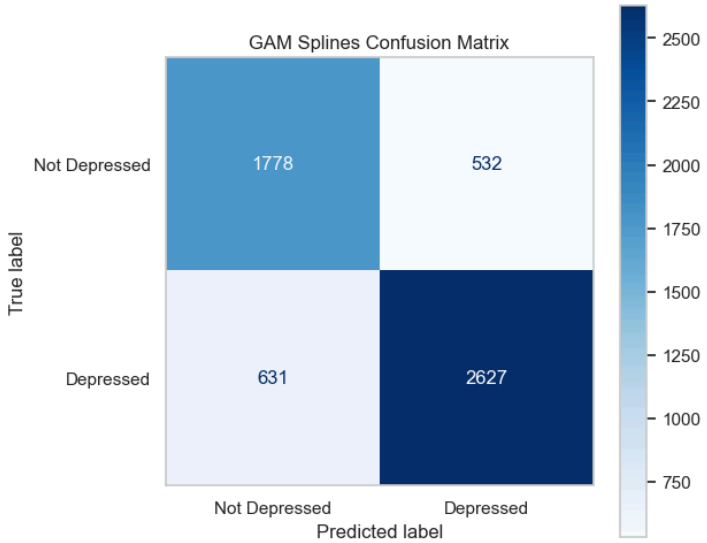






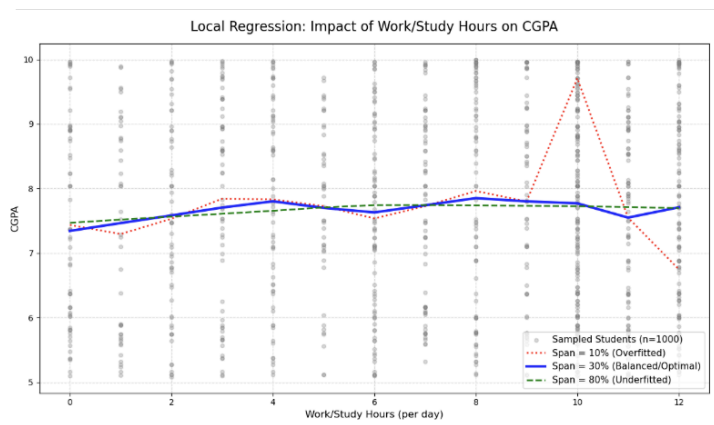
9.5 Visual Validation via Confusion Matrix

A graphical Confusion Matrix of our cross-validated predictions visually confirmed the real-world utility of the GAM. The model successfully identified 81% of actually depressed students (True Positives). While it produced a small number of False Positives, it heavily minimized False Negatives (missing a depressed student). This proves the model is acting optimally: for psychological screening, capturing true at-risk individuals is vastly more important than occasionally over-flagging healthy ones.



9.6 Local Regression

Local regression typically works the best when there are one or two continuous variables, in our case we use work/study hours to predict CGPA. In order for Local Regression to work, the data must be sorted along the x-axis, and the model will systematically move left to right, calculating a prediction for each point on the axis. The next step is to define the span, which is denoted as 'frac' in the code. This is responsible for determining the percentage of data points that will be included in that local neighborhood. We created three models, an overfitted model, an underfitted model, and a model in between, with 10%, 80%, 30% of neighbors, respectively. An important feature that Local Regression utilizes is weighting. It prioritizes students that are closer to the new student. At each point on the x-axis, the model performs a Weighted Least Squares (WLS) regression, which means it fits a line or a curve utilizing only the weighted neighbors. It moves across the whole axis repeating the process. Since doing this for every point would take a long time, we use a representative subsample of 1000 students. When graphing the three models the model with span = 30% most accurately represents that data. The graph shows that around 4 hours of studying is the best and it starts to plateau after that.



10. Tree-Based Methods

Professor Comment: "Bagging, random forests, boosting; good, gradient boosting is a plus!!"

We used tree-based methods to create a model that is highly interpretable, needs minimal data preparation, and is able to model very complex nonlinear relationships for classification.

10.1 Data Preparation

Unlike other models, trees require no feature scaling because trees evaluate each feature separately to find a place to split. Therefore, you are not required to use a StandardScaler, and you can directly pass numerical data into the model. Similarly to other models, though, trees need all inputs to be numbers, so we utilize One-hot encoding to turn categories into binary.

10.2 How the Tree Works

When using a tree-based method, the model looks at your data and decides what predictors to split on first. In order to do this, it looks at every variable and calculates the one that creates the group of students that is the “purest”(separetes healthy and depressed students the best).

- **Gini Impurity**: Calculates the probability of a random student being misclassified if they were randomly dropped into a branch. A score of 0 represents a perfectly pure node.
- **Entropy**: Measures the impurity within the dataset. It performs nearly the same as Gini, but typically is slightly slower. Generally, using Gini is the best approach.

10.3 Pruning

A huge problem with a single decision tree is that it will continue to grow without bound until every single student has their own leaf, completely memorizing the whole training data. This causes the model to perform very poorly on the test data.

In order to prevent this, you have to utilize Hyperparameters:

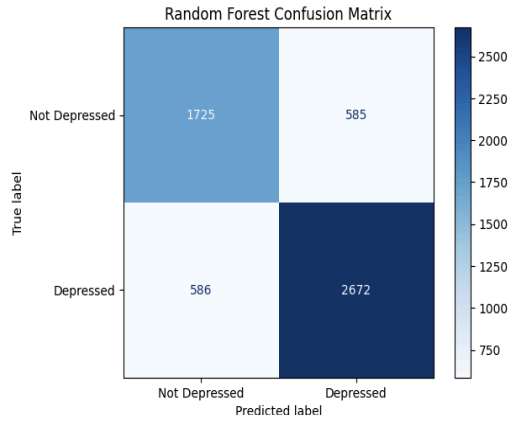
- max_depth: This is the maximum number of consecutive questions the tree can ask. We chose 4, so the tree is learning the general trend of the data and not the outliers.
- min_samples_split: This is the minimum number of students that can be present at a node before the tree can split again. We chose this parameter to be 50 to prevent making too many nodes.

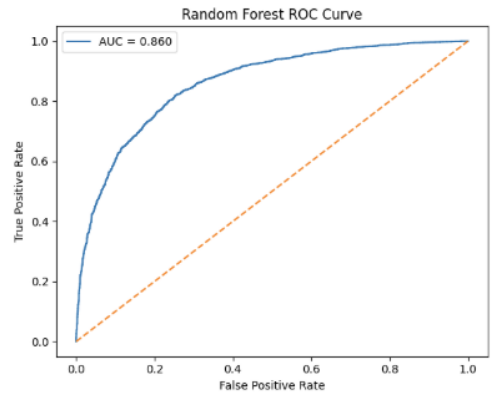
10.4 Ensembles

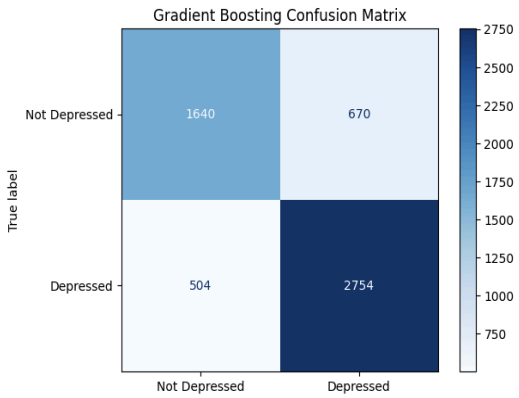
Most of the time, single decision trees are not good enough, so we need to use ensemble methods (Random Forests and Gradient Boosting) to combine hundreds or thousands of trees together.

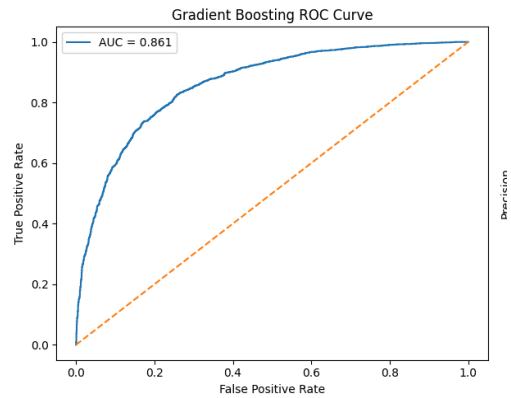
- Random Forests (Bagging): This model combines hundreds of independent, randomized trees together. Each tree is given a random subset of the data and of the predictors. As a result, this reduces the chances of overfitting and drastically reduces the variance. Then for classification, it is the majority vote of all of the trees. We built this model using 150 independent trees and the smoothing here occurs through bagging. Additionally, max_depth = 12, each tree was susceptible to overfitting but once all combined the variance leveled out creating a robust classifier.
- Gradient Boosting: The model builds each of the trees sequentially, building the next tree based off of the mistakes from the previous tree. The goal of boosting is to reduce the bias and achieve the highest possible classification rate on the testing data. For this model, 150 trees were built sequentially off of each other. Additionally, we used the smoothing parameter, shrinkage (Learning rate), of 0.1. This prevents any tree from making huge changes, scaling their contributions down, which forces the model to improve gradually.

(See evaluation figures on the next page)









12. Support Vector Machines

Professor Comment: “Be sure to use these”

The primary goal of Support Vector Machines (SVM) is to display all of the data and then construct the widest possible hyperplane between the depressed and healthy students. SVMs typically have high effectiveness with many predictors and robustness against overfitting. They utilize kernel functions to handle complex, non-linear data in order to maximize margin between classes.

12.1 Data Preparation

Since SVM is calculating distance between data points, it is required that you scale the data. For example, if two different variables have completely different ranges, with one range being greatly larger than the other, it will take priority and have a greater impact on the classification. Additionally, you must use One-hot encoding to convert all of the non-numerical categories into binary, so the SVM can plot each point.

12.2 Margins and Support Vectors

The main goal of SVM is to create a straight line (hyperplane) that separates the two classes, but it needs the best line to do so. The SVM model tries to maximize the margin on each side of the hyperplane. This margin is determined by points that are sitting on the edges of the hyperplane. These points are called the support vectors. The model relies so heavily on these support vectors that if you delete data far from the hyperplane, the model will not change. However, if a support vector is deleted, the model will look entirely different.

12.3 The Soft Margin

In a lot of data sets, it is impossible to completely separate the two classes. It is very possible for healthy students to be intermixed into a group of depressed students. If the SVM model uses a “Hard Margin”(no mistakes allowed) on a dataset like that, this will create a very jagged boundary line that overfits the training data. To control this, we use the hyperparameter C:

- High C (Strict): A high C will heavily penalize misclassifications. This creates a very narrow boundary line severely risking overfitting the training data.
- Low C (Soft): A low C means the model is more forgiving on misclassifications. This will create a wider, smoother boundary, which allows students to fall on the wrong sides in order to better represent the general trend of the data.

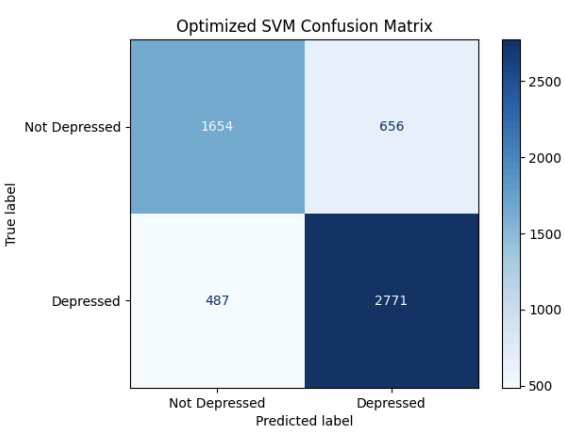
12.4 The Kernel Trick

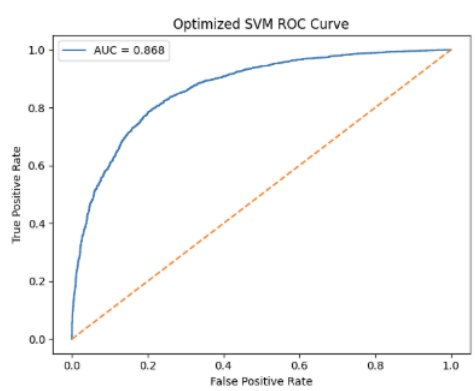
Most of the time, a straight boundary line will not do the trick. For example, say there is a ring of healthy students around a cluster of depressed students. To represent this, SVM utilizes the kernel trick, which projects the 2D data into higher dimension allowing for a flat sheet to be slid in between the two groups. Then when you project back to 2D it appears as a curve. During this tuning phase you will have to test different Kernels:

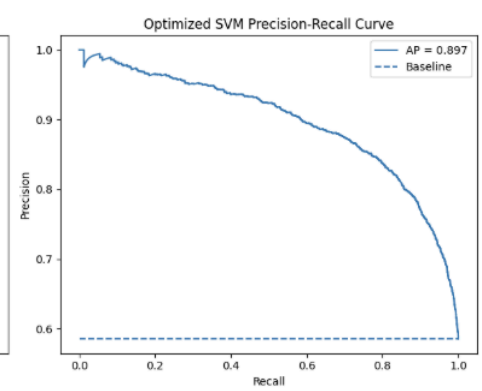
- Linear: This kernel draws a straight boundary between the groups
- Rbf (Radial Basis Function): This kernel draws organic, circular boundaries around the clusters of data.

12.5 Too much Data

SVMs are notoriously slow on large sets of data, and since our dataset has over 27,000 observations, we must take a small, representative sample of the data in order to test every combination of C and the type of kernel. We chose to use a sample size of 15%, and in doing so it determined a linear kernel is the best with a C value of 0.1.







Evaluation of Models

These models were primarily used for classification, with the exception of linear regression, which was used for prediction. We assessed the accuracy of the model prediction and classification accuracy by using test data, generated by 5-fold cross validation.

To compare the models, we used AUC-ROC where the model with the highest AUC was typically considered the best model. Additionally, we used confusion matrices to visualize where the different models were making their errors, as well as cross-validation to see which models performed consistently high across all folds. We also used the following metrics: Brier score, accuracy, precision, recall, F1 score, MCC, and AP.

Since we were predicting a medical condition, and there was class imbalance, we were careful about assessing the accuracy. A false negative was seen as significantly worse than a false positive, as it can be detrimental to the individual. Therefore, we used sensitivity (our ability to correctly predict students who actually have depression) and specificity (our ability to correctly predict students who do not have depression) to assess the accuracy of our predictions. It's important to maximize sensitivity because missing a depressed student, a false negative, carries a high real-world cost. Also we evaluated precision, which measures how many students who were flagged with depression were actually depressed. Using the F1 score, we found a balance between the precision and sensitivity ensuring our models did not overpredict one class.

Table 3: Model Performance Metrics on Test Set

Model Comparison — Test Set Performance

	ROC-AUC	Brier_Score	Accuracy	Precision	Recall	F1_Score	MCC	AP
Sparse (12 features) LR (No Suicide)	0.8678	0.1466	0.7892	0.7755	**0.9002**	**0.8332**	0.5623	**0.8968**
Deep Learning MLP	0.8674	0.1446	0.7935	0.8030	0.8573	0.8293	0.5706	0.8966
Optimized KNN	0.8565	0.1502	0.7906	0.7983	0.8591	0.8276	0.5643	0.8822
Decision Tree (Depth 4)	0.8266	0.1671	0.7672	0.8054	0.7940	0.7997	0.5220	0.8446
Random Forest	0.8601	0.1499	0.7897	0.8204	0.8201	0.8203	0.5669	0.8908
Gradient Boosting	0.8608	0.1484	0.7892	0.8043	0.8453	0.8243	0.5622	0.8892
LDA	**0.8680**	0.1444	0.7958	0.8034	0.8619	0.8316	0.5753	**0.8968**
QDA	0.8576	0.1551	0.7901	0.8009	0.8533	0.8263	0.5635	0.8863
Logistic GAM (No Suicide)	0.8676	0.1442	**0.7969**	0.8073	0.8576	0.8317	**0.5779**	0.8964
Baseline Linear SVM	0.8675	0.1442	0.7935	0.8071	0.8502	0.8281	0.5711	0.8966
Optimized SVM	0.8676	**0.1441**	0.7947	0.8086	0.8505	0.8290	0.5738	0.8967
PCR (No Suicide)	0.8625	0.1496	0.7868	**0.8303**	0.7990	0.8143	0.5649	0.8924
PLS-DA (No Suicide)	0.8669	0.1466	0.7909	0.8272	0.8125	0.8198	0.5711	0.8960

Project Summary

Summary of Analyses

After metrics for each model were compiled, we highlighted the model that scored highest for each metric in Table 3. Scores appeared highly consistent across all models. The sparse linear regression model had the greatest number of high scores, scoring highest out of all of the models on 3 of the 8 metrics - recall, F1, and AP. Due to its number of highest metric scores and the fact that it is comparable (less than 0.1 away) with the highest scoring models on the metrics we considered most important - ROC-AUC and accuracy. Again we would like to highlight the importance of these two measures as we would much rather our model predict a false positive than a false negative.

“What did you learn from this project?”

The linear regression model highlighted the bias versus variance tradeoff. The linear regression model lacks complexity and has little variance. It did not work well to fit a complex data set that had many influential ordinal variables. On the other hand, logistic regression, which is just slightly more flexible, was able to handle these variables outperformed all other models on several metrics. It showed a balance between bias and variance.

KNN was a good model to demonstrate the curse of dimensionality. With 15 variables creating 15 dimensions, volume expanded greatly. Even with 27,900 observations, the data points were sparse in the huge search space, and dimension reduction techniques like PCA improved model performance (and data visibility) greatly.

Both of these concepts demonstrate that more complex models are not always better. Throughout the project, shrinkage and dimension reduction techniques were applied to simplify models, often improving model performance, and providing more efficient, parsimonious models. That doesn’t mean that simple models always win out over complex models. Simplifying models too much can cause underfitting and other issues. There is a fine balance between simplicity and complexity that must be reached for optimal model performance.

It is also important to take the time and effort to test and refine models. As could be seen when shrinkage and dimension reduction techniques were applied, the first model can almost always be improved. Even in models where such techniques are not applied but the data set contains many predictors, being able to eliminate insignificant predictors can benefit the model. Further, we learned that it is important to evaluate models on test data and not training data. The model should always perform well on the data on which it is being trained. We are interested in how it performs on data it is not familiar with. For this reason, all evaluations and results were taken only on the test data, to truly demonstrate our model’s effectiveness.

Professor Comments/Questions

Although all of these questions/comments have been addressed in the sections above, for clarity we are including all remaining professor feedback/comments that did not apply to a specific model/section:

- *"Will you be using the trained model from supervised learning for prediction or classification?"*
    - **This was clearly addressed, as all but one of our models demonstrated binary classification for the target variable, "Depression", which is a categorical. Thus, this is a classification process.**
- *"How will you assess the accuracy of your predictions or classifications?"*
    - Professor Comment: "Using the test data is the best"
    - **Addressed in the [Evaluation of Models]() section, we used a variety of metrics and evaluations on the held-out test data to assess the accuracy and usefulness of each of our trained models.**
- *"How will you compare the models you consider?"*
    - Professor Comments:
        - "Need to consider training, validation and testing data sets; need to do this for neural nets"
        - "Cross-validation for some models Cp, BIC, AIC, Adjusted R^2; need to use this for linear model selection"
        - "Confusion matrices for the test data; need to use these for comparing all of the different methods for classification like was done in the Chapter 4 homework"
        - "Use of the F1 score is a plus; using AUC-ROC where the model with the highest AUC is typically considered the best model is also a plus"
    - **As shown in each of their respective sections, we addressed each of these comments: train/test/validate sets were created, most models used 5-fold cross-validation to evaluate generalizeability, confusion matrices are provided for all classifier models, and we used F1 score, AUC-ROC, and many other metrics for our final evaluations.**

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